

Homework #3. Exploratory Data Analysis

Context

It's assumed that your dataset has already been downloaded and prepared during Homework #2.

Now, you take the role of a data analyst who received this dataset and must explore it from multiple angles - to understand what it contains, what stories it tells, and what insights it hides.

Your task is to perform exploratory data analysis (EDA) - to describe, visualize, and interpret patterns, trends, and relationships in your data. You are expected to ask many questions, search for answers in the data, and use visualization to communicate your findings. The goal is to transform a raw dataset into a clear understanding of its structure, dynamics, and potential for further analysis.

Task #1

Create a new Jupyter Notebook named `eda.ipynb`. Record the time you spent working on this homework in one of the first cells.

Perform an exploratory data analysis (EDA) on the Reddit dataset you collected in Homework #2.

Your task is to explore the data from different perspectives - temporal, structural, and textual - to understand what is happening inside your dataset and what insights it can reveal.

Focus on describing, visualizing, and interpreting key patterns, distributions, relationships, and trends.

If you have both **submissions** and **comments** data, analyze them separately and, if appropriate, merge them to observe combined dynamics. When merging, briefly explain how and why you did it.

The **report itself should be written and visualized inside your Jupyter Notebook**. Use text cells for explanations and Markdown to describe your findings. The notebook should be self-contained and readable as a complete report.

When finished, export your notebook to a **PDF file** for submission.

Evaluation and Expectations

Sometimes people try to choose the shortest path - doing only the minimum required. However, exploratory analysis is not about checking boxes but about curiosity and persistence. Your work will be evaluated relative to the best submissions in the group. Roughly five of the most comprehensive and insightful works will define the top quality benchmark (100%), and other projects will be graded proportionally to the amount and depth of analysis completed. You are expected to ask questions and use your dataset to answer them through analysis and visualization.

A good EDA typically involves **at least 20 meaningful** visualizations - ranging from simple descriptive ones (e.g. "Which subreddits are most active?") to more complex, interpretive ones (e.g. "How does participation in certain subreddits change during specific events?").

Your exploration should be mostly visual. Each visualization must be clear, titled, and properly labeled:

- The title explains what the figure shows.
- The x- and y-axes have meaningful names and units.
- The legend, if present, clearly describes all elements.
- Avoid unnecessary decoration - clarity and insight matter most.

I expect you to go beyond surface-level plots (like a few counts or histograms). Think about patterns in time, relationships between variables, anomalies, or trends that might surprise you. Try to explain why you see these patterns.

This work is designed to strengthen your analytical thinking, creativity, and your ability to tell a story with data. The more you explore, the more insights you will find.

Recommendations

Start by revisiting **Lecture #3 (Dataviz)** and review the purpose and logic of exploratory analysis.

You can also check this short overview:

https://en.wikipedia.org/wiki/Exploratory_data_analysis

Think of your work as a guided exploration. Begin with simple questions and gradually move to more specific or complex ones. For example:

- How many records do I have in total and per month?
- What are the most active subreddits?
- How does user activity change over time?
- Which words or topics dominate certain periods or communities?
- Are there noticeable anomalies or peaks, and what might explain them?

Structure your notebook as a clear story:

1. **Overview:** dataset shape, number of records, timeframe, basic info.

2. **Activity and distribution:** counts by month, subreddit, or author.
3. **Content analysis:** frequent terms, topics, or sentiment (if relevant).
4. **Interesting findings:** correlations, spikes, or patterns worth explaining.

Use visualizations to drive insight. Try to choose the right type of chart for the data you explore. For visual inspiration and best practices, review:

- <https://www.data-to-viz.com/>
- <https://datavizproject.com/>
- <https://datavizcatalogue.com/>
- <https://github.com/Financial-Times/chart-doctor/tree/main/visual-vocabulary>

Color and design tools that can help:

- <https://projects.susielu.com/viz-palette>
- <https://colorbrewer2.org/#type=sequential&scheme=BuGn&n=3>
- <https://learnui.design/tools/data-color-picker.html#palette>

Avoid meaningless visualizations like word clouds.

Focus instead on charts that explain relationships and show patterns in context.

Finally, think about your findings in a **wider context**.

If you notice peaks, shifts, or sudden changes in discussion intensity, check what was happening in the world, in your country, or in specific regions at that time.

Understanding how real events influence online behavior will help you interpret your results and give your analysis a deeper, more meaningful perspective.

Intermediate results presentation

06/10/2026, 23:59 (5pt)

Expected Outcome

1. PDF report (file named “3_0_<team_id>.pdf”) exported from the file `eda.ipynb`. This report should consist of your (some) exploration for datasets, a list of questions (assumptions, hypothesis, ideas to test) you have to the data, and some drawings of basic information (distributions, etc).
It's just a preliminary report, I'm going to score the fact that you've submitted a file with the information above but with remaining work (not finalized) to be done during the second week of work.

Final Deadline

11/10/2026, 23:59 (max: 15pt + extra)

Expected Outcome

1. PDF report (file named “3_<team_id>.pdf”) exported from the file `eda.ipynb` with the well-structured report, with storytelling, drawings, questions, and answers (in the visual form mostly, but not only).
2. 1 csv file named “3_<team_id>.csv” with all fields filled.

ATTENTION: naming is important.

Your team ID can be found here -

https://docs.google.com/spreadsheets/d/1TCuefc_Rsx1jWlv_wViAZVQ7N1w_WBlt/edit?usp=sharing&oid=104355185867688082261&rtpof=true&sd=true

Materials

Lecture #3

Some basic tools (you can apply to the data)

- Text Mining: Word Relationships
https://uc-r.github.io/word_relationships
- Count basic statistics (Basics of Text Mining in R - Bag of Words)
http://rstudio-pubs-static.s3.amazonaws.com/256588_57b585da6c054349825cba46685d8464.html

Visual ideas

1. <https://www.data-to-viz.com/>
2. <https://datavizproject.com/#>
3. <https://datavizcatalogue.com/>
4. <https://github.com/Financial-Times/chart-doctor/tree/main/visual-vocabulary>

Colours

1. <https://projects.susielu.com/viz-palette>
2. <https://colorbrewer2.org/#type=sequential&scheme=BuGn&n=3>
3. <https://learnui.design/tools/data-color-picker.html#palette>

ggplot2

1. GG plot usage
<http://r-statistics.co/Top50-Ggplot2-Visualizations-MasterList-R-Code.html>

2. Themes

<https://github.com/jrnold/ggthemes>

<https://github.com/Mikata-Project/ggthemr>

GEO

1. <https://kepler.gl/#/>
<https://github.com/keplergl/kepler.gl>
2. <https://datashader.org/>
<https://github.com/holoviz/datashader/>
3. <https://geoffboeing.com/2016/11/osmnx-python-street-networks/>
<https://github.com/gboeing/osmnx>
4. <https://github.com/python-visualization/folium>
5. <https://github.com/karimbahgat/PyGeoJ>
6. <https://github.com/riatelab/linemap>

List of tools

1. <https://docs.google.com/spreadsheets/d/1miN1dUvPMmnLdhunhZCil80LwlJ3xj8gRXrUNA-KrIE/edit#gid=0>

Dataviz ideas

1. <https://pudding.cool/>
2. <https://exhibits.stanford.edu/dataviz>
3. <https://setosa.io/#/>
4. <http://czrt.by/projects/>
5. <https://www.makeovermonday.co.uk/makeovers/makeovers-2018/>
6. <https://www.bloomberg.com/graphics/2015-whats-warming-the-world/>
7. <https://www.visualisingdata.com/resources/>